G15 Connectors for internal tubes

One-stop connection solution

Low pressure test | Easy to replace | Automatic operation | Full range of models| Quick operation | Diversified seal materials | Small size | Internal expansion for sealing

The G15 series pneumatic quick seal connector utilizes air pressure to inflate the seal and create an internal airtight connection with virtually any type of tube, which is ideal for high volume automated production testing.



High-efficient automatic connection solution

* Pneumatic actuated, providing more stable pilot pressure
* Instantaneous connection, suitable for smooth, rough, porous, threaded, and various irregular fittings
* Seals material, exquisite and reliable, causing no damage to the product surface
* Available for gripping-positioning applications for robotic manipulators, simple maintenance and replaceable seals, which significantly extends the service life of the connectors

Features of G15 series quick connectors

1. Pneumatic actuated

Air can be injected through the rear pilot hole to push the built-in piston and compress the seal to achieve tight sealing.

2. Automatic operation

Through the two mounting holes, the connector can be fixed to any actuator, realizing automatic operation with accuracy and reliability.

3. Straight-through test port

This port reduces the intermediate part, improves the sealing and keeps the flow high.

4. Stroke limiter

The build-in stroke limiter limits the maximum position of the piston, and prevents the seal from falling out due to overload.

G15 series quick connectors are available in multiple body sizes, ranging from 0 to 5, offering a sealing range of 8.5mm to 44mm.

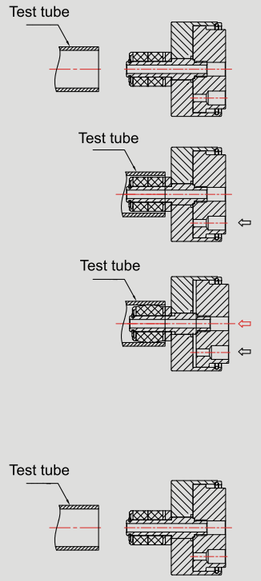
Body sizes



G15 Connectors for internal tubes

One-stop connection solution

Low pressure test | Easy to replace | Automatic operation | Full range of models| Quick operation | Diversified seal materials | Small size | Internal expansion for sealing



Enable easier and reliable connections

Step 1:

Insert the test tube into test port of the G15 series, and secure the test tube and connector.

Step 2:

Introduce pilot pressure to pilot port. The seal is squeezed, creating expansion, which in turn clings to inner wall of test tube. In result, a leak tight seal is formed.

Step 3:

a. Filling the test port with test medium upon the formation of the seal (such as compressed air, test fluid, etc.)

b. After the test is completed, release the inner test pressure to atmospheric pressure, then remove the pilot pressure, the seal is no longer be squeezed, the seal connection release the test tube.

Step 4:

Once the above operations are completed, the test tube can be removed from the connector.

G15 quick connectors



G15 Connectors for internal tubes

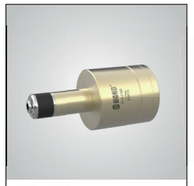
One-stop connection solution

Low pressure test | Easy to replace | Automatic operation | Full range of models| Quick operation | Diversified seal materials | Small size | Internal expansion for sealing

Advantages of G15 quick connectors



Pneumatic actuated: Good sealing and simple operation



Flexible customization: Flexible clamping position to avoid space collisions



Quick seals replacement: Simple maintenance & extended service life



Compact size: Less operating space occupied & can be placed anywhere in the fixture



Automatic operation: Higher efficiency & more consistent test results

Specifications of G15 quick connectors

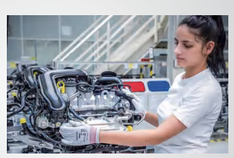
Max operation pressure	From Vacuum to 8 Bar
Sealing range	8.5mm - 44mm
Minimize insert depth	BS0, BS1:15mm BS3, BS4: 25mm
Termination port	1/8"BSPP,3/8"BSPP
Material of main body	Aluminum alloy
Main seal	Neoprene/silicone
Internal seal	Buna-N
Pilot pressure	4-6 Bar
Operation temperature	-20℃ - 120℃

The G15 system is designed to create a leak-free internal connection for a wide range of fittings by injecting the pilot pressure into the tail drive hole and forcing the piston forward, thereby squeezing the seal at the front end. G15 connectors are the perfect choice for high-volume automated production applications.

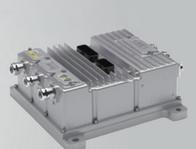
Test on cooling tube
Turbo
Radiator
Plastic port
Fuel pump



Cooling tube in electric motor



Engine assembly test

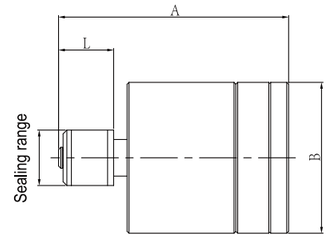


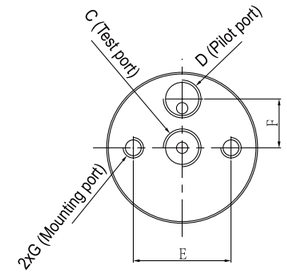
Motor controller

G15 Connectors for internal tubes

One-stop connection solution

Low pressure test | Easy to replace | Automatic operation | Full range of models| Quick operation | Diversified seal materials | Small size | Internal expansion for sealing





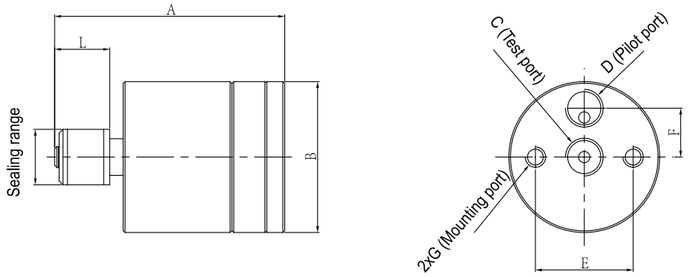
Part Number	Sealing Range	Dim	Dim	Dim	Dim D	Dim	Dim	Dim G	Dim L	Main Seals Package**
G15-0080	8.0 - 8.5	50.4	33.5	M5	M5	20.3	10	M5	12.2	SG150080
G15-0085	8.5 - 9.5									SG150085
G15-0095	9.5 - 10.5									SG150095
G15-1100	10.5 - 11.5	61.8	40	1/8"BSPP	1/8"BSPP	25.9	13	M6	14.5	SG151100
G15-1110	11.5 - 12.5									SG151110
G15-1120	12.5 - 13.5									SG151120
G15-1130	13.5 - 14.5									SG151130
G15-1140	14.5 - 15.5									SG151140
G15-1150	15.5 - 17.0									SG151150
G15-2160	17.0 - 18.0	61.8	40	1/8"BSPP	1/8"BSPP	25.9	13	M6	14.5	SG152160
G15-2170	18.0 - 19.0									SG152170
G15-2180	19.0 - 20.0									SG152180
G15-2190	20.0 - 21.0									SG152190
G15-2200	21.0 - 22.0									SG152200
G15-2210	22.0 - 23.0									SG152210
G15-3220	23.0 - 24.0	63.6	60	1/8"BSPP	1/8"BSPP	40.6	18.3	M6	24.5	SG153220
G15-3230	24.0 - 25.0									SG153230
G15-3240	25.0 - 26.0									SG153240
G15-3250	26.0 - 27.0									SG153250
G15-3260	27.0 - 28.0									SG153260
G15-3270	28.0 - 29.0									SG153270
G15-4280	29.0 - 30.0	63.6	60	1/8"BSPP	1/8"BSPP	40.6	18.3	M6	24.5	SG154280
G15-4290	30.0 - 31.0									SG154290
G15-4300	31.0 - 32.0									SG154300
G15-4310	32.0 - 33.0									SG154310
G15-4320	33.0 - 34.0									SG154320
G15-4330	34.0 - 35.0									SG154330
G15-5340	35.0 - 37.0	93.5	88.5	3/8"BSPP	1/8"BSPP	58.7	29.4	M8	31.3	SG155340
G15-5350	36.0 - 38.0									SG155350
G15-5370	38.0 - 40.0									SG155370
G15-5390	40.0 - 42.0									SG155390
G15-5400	41.0 - 43.0									SG155400
G15-5420	43.0 - 45.0									SG155420

1. Nonstandard models of G15 quick connectors include: larger dimensions (up to 80mm), longer center rod, larger OD of the termination port, shorter seal thickness, etc.
2. Customized models of G15 quick connectors include: sealing various threads, smaller volumes, complex shapes, etc. Please contact your local distributor or GripSeal sales representative for more information.

G15RC Connectors for internal tubes (BSPT threads)

One-stop connection solution

Low pressure test | Easy to replace | Automatic operation | Full range of models| Quick operation | Diversified seal materials | Small size | Internal expansion for sealing



Part Number	Sealing Range	Dim	Dim B	Dim C	Dim	Dim	Dim F	Dim G	Dim L	Main Seals Package**
G15RC-1100	10.5 - 11.5	61.8	40	1/8"BSPT		25.9	13	M6	14.5	SG151100
G15RC-1110	11.5 - 12.5									SG151110
G15RC-1120	12.5 - 13.5									SG151120
G15RC-1130	13.5 - 14.5									SG151130
G15RC-1140	14.5 - 15.5									SG151140
G15RC-1150	15.5 - 17.0									SG151150
G15RC-2160	17.0 - 18.0	61.8	40	1/8"BSPT		25.9	13	M6	14.5	SG152160
G15RC-2170	18.0 - 19.0									SG152170
G15RC-2180	19.0 - 20.0									SG152180
G15RC-2190	20.0 - 21.0									SG152190
G15RC-2200	21.0 - 22.0									SG152200
G15RC-2210	22.0 - 23.0									SG152210
G15RC-3220	23.0 - 24.0	63.6	60	1/4"BSPT	1/8"BSPT	40.6	18.3	M6	24.5	SG153220
G15RC-3230	24.0 - 25.0									SG153230
G15RC-3240	25.0 - 26.0									SG153240
G15RC-3250	26.0 - 27.0									SG153250
G15RC-3260	27.0 - 28.0									SG153260
G15RC-3270	28.0 - 29.0									SG153270
G15RC-4280	29.0 - 30.0	63.6	60	1/4"BSPT	1/8"BSPT	40.6	18.3	M6	24.5	SG154280
G15RC-4290	30.0 - 31.0									SG154290
G15RC-4300	31.0 - 32.0									SG154300
G15RC-4310	32.0 - 33.0									SG154310
G15RC-4320	33.0 - 34.0									SG154320
G15RC-4330	34.0 - 35.0									SG154330
G15RC-1120-250*	13.0 - 14.0	86.8	40	1/8"BSPT	1/8"BSPT	25.9	13	M6	14.5	SG151120
G15RC-2180-500*	19.0 - 20.0	111.8								SG152180
G15RC-3260-750*	27.0 - 28.0	138.6	60	1/4"BSPT		40.6	18.3		24.5	SG153260
G15RC-4280-500*	29.0 - 30.0	113.6								SG154280

Note: The asterisk (*) means that extended shaft versions are available, where H is an integer multiple of 25 mm, and the maximum value of H is 150 mm.
1. Nonstandard models of G15RC quick connectors include: larger dimensions (up to 80mm), longer center rod, larger OD of the termination port, shorter seal thickness, etc.
2. Customized models of G15RC quick connectors include: sealing various threads, smaller volumes, complex shapes, etc. Please contact your local distributor or GripSeal sales representative for more information.

G15 Pro Connectors for internal tubes One-stop connection solution

Sensor feedback | Identity confirm after connector in place | Internal sealing | Easy to maintain | Full range of models | Automatic operation

In comparison to the traditional G15 series connectors, G15 Pro series can be mounted externally with a magnetic sensor, which enables more accurate tracking of the connector drive, seal expansion, eliminating incorrect connections resulted from improper sealing, and enhancing the efficiency of automated inspection.



High-efficient automatic connection solution

* Pneumatic actuated, providing more stable pilot pressure
* Instantaneous connection, suitable for smooth, rough, porous, threaded, and various irregular fittings
* Seals material, exquisite and reliable, causing no damage to the product surface
* Available for gripping-positioning applications for robotic manipulators, simple maintenance and replaceable seals, which significantly extends the service life of the connectors

Features of G15 Pro series quick connectors

1. Sensor feedback

Verify whether the connection good or not by setting the position of the sensor.

2. Pneumatic actuated

Air can be injected through the rear pilot hole to push the built-in piston and compress the seal to achieve tight sealing.

3. Automatic operation

Through the two mounting holes, the connector can be fixed to any actuator, realizing automatic operation with accuracy and reliability.

4. Damage-free connection

The seal is clamped without causing any damage to the surface of the test piece.

G15 Pro series quick connectors are available in multiple body sizes, ranging from 0 to 4, offering a sealing range of 8.5mm to 35mm.

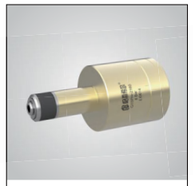
Body sizes



G15 Pro Connectors for internal tubes One-stop connection solution

Sensor feedback | Identity confirm after connector in place | Internal sealing | Easy to maintain |
Full range of models | Automatic operation

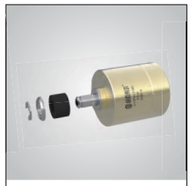
Advantages of G15 Pro quick connectors



Flexible customization, avoiding space collisions



Position sensor: Reporting the movement of the internal piston in real time



Quick seals replacement: Simple maintenance & extended service life

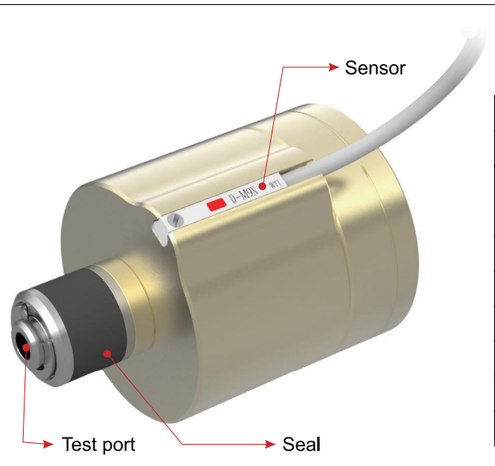


Different materials of seals to meet diversified customer needs



Automatic operation: Higher efficiency & more consistent test results

Specifications of G15 Pro quick connectors



Max operation pressure	From Vacuum to 8 Bar
Sealing range	8.5mm - 35mm
Minimize insert depth	BS1, BS2:15mm BS3, BS4: 25mm
Termination port	1/8"BSPP
Material of main body	Aluminum alloy
Main seal	Neoprene/silicone
Internal seal	Buna-N
Pilot pressure	4-6 Bar
Operation temperature	-20℃ - 120℃

The G15 Pro system is designed to create a leak-free internal connection for a wide range of fittings by injecting the pilot pressure into the tail drive hole and forcing the piston forward, thereby squeezing the seal at the front end. G15 Pro connectors are the perfect choice for high-volume automated production applications.

Test on cooling tube
Turbo
Radiator
Plastic port
Fuel pump



Cooling tube in electric motor



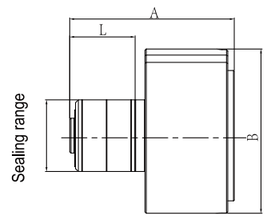
Engine assembly test

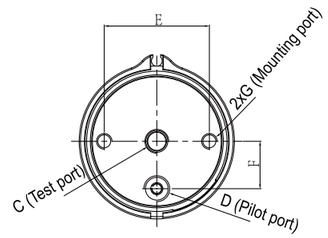


Motor controller

G15 Pro Connectors for internal tubes One-stop connection solution

Sensor feedback | Identity confirm after connector in place | Internal sealing | Easy to maintain |
Full range of models | Automatic operation





Part Number	Sealing Range	Dim A	Dim B	Dim C	Dim D	Dim E	Dim F	Dim G	Dim L	Main Seals Package**
G15Pro-0080	8.0 - 8.5	54.7	38	M5	M5	20.3	10	M5	12.2	SG150080
G15Pro-0085	8.5 - 9.5									SG150085
G15Pro-0095	9.5 - 10.5									SG150095
G15Pro-1100	10.5 - 11.5	62	45	1/8"BSPP		25.9	13	M6	14.5	SG151100
G15Pro-1110	11.5 - 12.5									SG151110
G15Pro-1120	12.5 - 13.5									SG151120
G15Pro-1130	13.5 - 14.5									SG151130
G15Pro-1140	14.5 - 15.5									SG151140
G15Pro-1150	15.5 - 17.0									SG151150
G15Pro-2160	17.0 - 18.0	62	45	1/8"BSPP		25.9	13	M6	14.5	SG152160
G15Pro-2170	18.0 - 19.0									SG152170
G15Pro-2180	19.0 - 20.0									SG152180
G15Pro-2190	20.0 - 21.0									SG152190
G15Pro-2200	21.0 - 22.0									SG152200
G15Pro-2210	22.0 - 23.0									SG152210
G15Pro-3220	23.0 - 24.0	63.6	64	1/8"BSPP		40.6	18.3	M6	24.5	SG153220
G15Pro-3230	24.0 - 25.0									SG153230
G15Pro-3240	25.0 - 26.0									SG153240
G15Pro-3250	26.0 - 27.0									SG153250
G15Pro-3260	27.0 - 28.0									SG153260
G15Pro-3270	28.0 - 29.0									SG153270
G15Pro-4280	29.0 - 30.0	63.6	64	1/8"BSPP		40.6	18.3	M6	24.5	SG154280
G15Pro-4290	30.0 - 31.0									SG154290
G15Pro-4300	31.0 - 32.0									SG154300
G15Pro-4310	32.0 - 33.0									SG154310
G15Pro-4320	33.0 - 34.0									SG154320
G15Pro-4330	34.0 - 35.0									SG154330

1. Nonstandard models of G15 Pro quick connectors include: larger dimensions (up to 80mm), longer center rod, larger OD of the termination port, shorter seal thickness, etc.
2. Customized models of G15 Pro quick connectors include: sealing various threads, smaller sizes, complex shapes, etc.
Please contact your local distributor or GripSeal sales representative for more information.

G15K Pneumatic internal expansion connectors

One-stop connection solution

Fast replacement | Automation | Internal expansion | Compact size | Diversified seal materials

The G15K series pneumatic quick seal connector utilizes air pressure to inflate the seal and create an internal airtight connection with virtually any type of tube, which is ideal for high volume automated production testing. The innovative "quick maintenance" technology enables the replacement of seals without disassembling the connector, greatly shortening the maintenance time of the connector.



Pneumatic internal expansion connectors

* Reducing overall manufacturing costs through more efficient testing
* Automated testing to maximize throughput capabilities
* Improving the test stability
* Fast replacement of seals to shorten the maintenance time

Features of G15K series quick connectors

1. Quick maintenance

The button can be pressed to quickly remove the seal, greatly shortening the maintenance time.

2. Sealing through internal expansion

The internal structure of the connector is compact, with a small volume and a wide range of applications.

3. Damage-free connection

The parts in contact with the test port are all rubber parts, which will not cause damage to the customer's products.

4. Pneumatic actuated

The expansion and contraction of the seal can be controlled by air pressure, and the product is suitable for automation application scenarios.

G15K series quick connectors are available in multiple body sizes, ranging from 0 to 4, offering a sealing range of 8.5mm to 35mm.

Body sizes

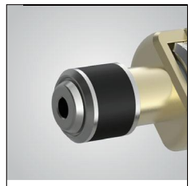


G15K Pneumatic internal expansion connectors

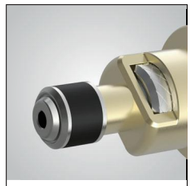
One-stop connection solution

Fast replacement | Automation | Internal expansion | Compact size | Diversified seal materials

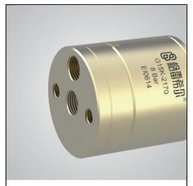
Advantages of G15K quick connectors



Sealing through internal expansion: Seals expanded to achieve sealing



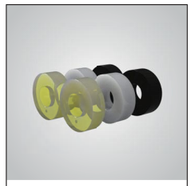
Fast replacement: Fast replacement of the seal to reduce the maintenance time and costs



Standard threaded ports: Standard threaded pilot, test, and mounting ports, facilitating selection of accessories

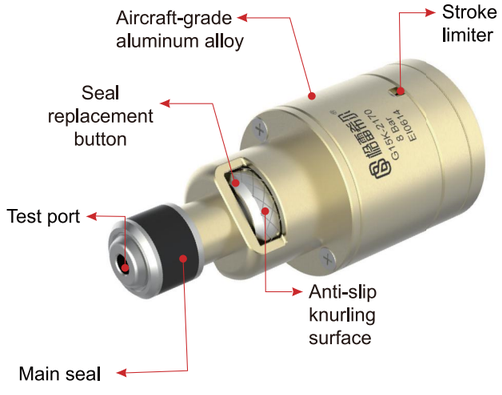


Compact size: Compact internal structure for a small size and a wide range of applications



Diversified materials: Diversified seal materials to meet requirements of different test scenarios.

Specifications of G15K quick connectors



Max operation pressure	From Vacuum to 8 Bar
Sealing range	8.5mm - 35mm
Minimize insert depth	BS1, BS2:15mm BS3, BS4: 25mm
Termination port	M5,1/8"BSPP
Material of main body	Aluminum alloy
Main seal	Neoprene/polyurethane/silicone
Internal seal	Buna-N
Pilot pressure	4-6 Bar
Operation temperature	-20℃ - 120℃

The G15K series connectors have the advantages of quick maintenance, damage-free connection, and no leakage, making them the best connection solution for the automated pressure test, running test, flow test, etc.

Seal replacement procedure:



Step 1: Press the silver button and the front end will automatically pop up **(it is recommended to cover the front end with your palm to prevent the component from popping up and scattering).** Then remove the old seal.



Step 2: Put on the new seal.



Step 3: Press in the new seal until an in-place sound is heard.